

⚡ WattWatch: Analyzing Urban Energy Consumption for Smarter Cities

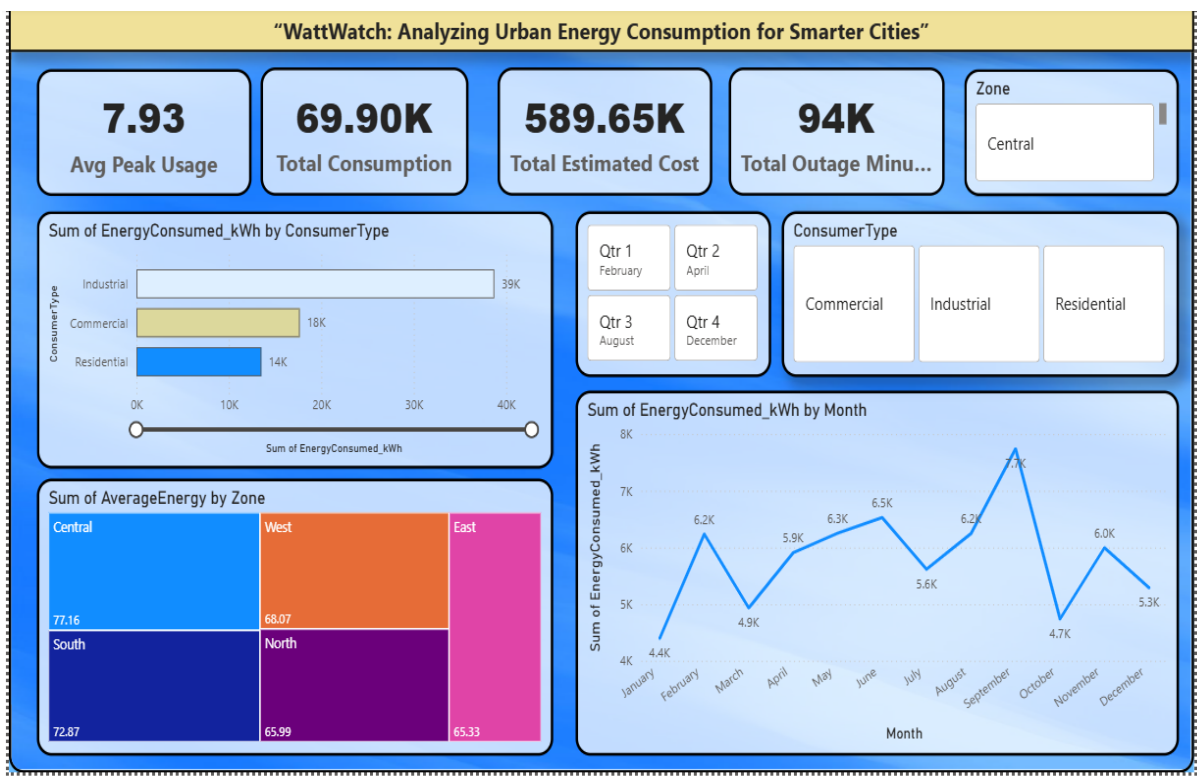
📖 Project Overview

WattWatch is a Smart City Energy Analytics project developed using SQL Server and Power BI. The project analyzes electricity consumption patterns across different city zones, identifies energy hotspots, monitors outage trends, and generates actionable insights for smarter urban planning.

The objective is to help municipal authorities understand energy usage behavior, improve infrastructure reliability, and support sustainable energy management.

📅 Project Duration

3 Hours



Project Brief

A municipal data team wants to assess energy usage across various zones of a smart city. The goal is to uncover trends in electricity consumption, identify energy wastage, detect outage-prone areas, and provide data-driven recommendations for sustainability.

This project combines:

- SQL-based Data Analysis
- Power BI Dashboard Development
- Business Insight Generation

Dataset Information

Dataset Name: SmartCityEnergy.csv

Dataset Schema

Column Name	Description
MeterID	Unique identifier for each smart meter
Zone	City zone (North, South, East, West, Central)
ConsumerType	Residential / Commercial / Industrial
Date	Reading date
EnergyConsumed_kWh	Total kilowatt-hours consumed
PeakUsage_kWh	Maximum kWh drawn in a single hour
OutageMinutes	Power outage duration in minutes
MeterStatus	Active / Faulty
TariffRate	Price per kWh at the time of billing

Project Objectives

- Analyze energy consumption across city zones.
 - Identify high-energy-consuming consumers.
 - Monitor meter faults and outage patterns.
 - Detect inefficient energy usage.
 - Analyze monthly consumption trends.
 - Compare weekday and weekend peak usage.
 - Build an interactive Power BI dashboard.
 - Generate actionable recommendations for city planners.
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SQL Analysis

The following analyses were performed using SQL Server:

1 Total & Average Energy Consumption by Zone

- Calculate total energy consumption for each zone.
- Calculate average energy consumption for each zone.

2 Top 5 Highest Energy Consuming Consumers

- Identify consumers with the highest electricity usage.

3 Monthly Consumption Trends

- Analyze monthly electricity consumption patterns across zones.

4 Average Cost by Zone

Formula used:

$\text{EnergyConsumed_kWh} \times \text{TariffRate}$

5 Fault & Outage Analysis

- Identify faulty meters.
- Detect meters with high outage durations.

6 Energy Efficiency Analysis

- Detect zones with high consumption and frequent outages.

7 Weekday vs Weekend Peak Usage

- Compare electricity demand during weekdays and weekends.

Power BI Dashboard

An interactive Power BI dashboard was developed to visualize key energy metrics and trends.

Dashboard Components

KPI Cards

- Total Consumption
- Average Peak Usage
- Total Outage Minutes
- Total Estimated Cost

Energy Usage by Zone

Treemap visualization showing electricity distribution across city zones.

Monthly Consumption Trend

Line chart displaying monthly electricity consumption patterns.

Energy Usage by Consumer Type

Bar chart comparing:

- Residential
- Commercial
- Industrial

consumers.

Interactive Slicers

- Zone
- Consumer Type
- Quarter
- Date Range

Meter Health Monitoring

Conditional formatting highlights:

- Active Meters
- Faulty Meters

Dashboard Preview

Place your dashboard screenshot inside the **images** folder and name it **dashboard.png**

Key Findings

Energy Consumption

- Industrial consumers account for the highest energy usage.
- Central zone records the highest average energy consumption.
- Residential consumers show relatively stable demand throughout the year.

Monthly Trends

- Energy consumption varies across months, indicating seasonal demand patterns.
- Peak demand occurs during specific months and requires capacity planning.

Reliability Analysis

- Certain zones experience higher outage durations than others.
- Faulty meters impact monitoring accuracy and operational efficiency.

Energy Efficiency

- High consumption combined with frequent outages indicates inefficient zones.
- Infrastructure upgrades may significantly improve energy performance.

Recommendations

- Implement energy conservation programs in high-demand zones.
- Prioritize maintenance in outage-prone areas.
- Increase monitoring of industrial consumers.
- Deploy predictive maintenance using smart meter data.

⌘ Technologies Used

Technology	Purpose
SQL Server	Data Storage & Analysis
SSMS	Query Development
Power BI Desktop	Dashboard & Visualization
Microsoft Excel	Data Export
GitHub	Version Control

📁 Project Structure

WattWatch/
|

```
├── SmartCityEnergy.csv
├── SmartCity_SQL.sql
├── WattWatchDashboard.pbix
├── EnergyFindings.txt
├── README.md
└── images/
    └── dashboard.png
```

Deliverables

SmartCity_SQL.sql

Contains all SQL queries used for analysis.

WattWatchDashboard.pbix

Interactive Power BI dashboard.

EnergyFindings.txt

Contains business insights derived from the analysis.

SmartCityEnergy.csv

Dataset used for analysis.

Dashboard KPIs

KPI	Description
Total Consumption	Total energy consumed
Avg Peak Usage	Average peak demand
Total Outage Minutes	Total outage duration
Total Estimated Cost	Estimated electricity cost

Learning Outcomes

This project demonstrates:

- SQL Data Analysis
- Data Aggregation & Reporting
- Power BI Dashboard Development

- Business Intelligence
 - Data Visualization
 - Smart City Analytics
 - KPI Monitoring
 - Interactive Reporting
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